

Sharper foundation + connected methods v1.3.2

oocrprep.py \$1.3/1.8

- deskew (8° → 0)
- denoise · Otsu binarize
- orientation (OSD)
- better OCR input

PRE-OCR

olayout.py \$2.4

- title/header/para/caption
- header/footer/figure
- from PyMuPDF blocks (no ML)
- reading order · /api/layout

LAYOUT

odedup.py \$7.1

- word-shingle fingerprints
- Jaccard clustering
- same TM diff change# → cluster
- prefer latest · de-dupe hits

EDITIONS

ocrossval.py \$7.5

- measures + tables + IETM agree
- confidence up (3-way=1.0)
- disagreements → conflict flag
- feeds Masterfile confidence

AGREEMENT

R12 progress — foundation & cross-document lanes done

o now spans \$1.3/1.8 (OCR prep), \$2.2/2.3/2.4/2.6 (tables + layout + cleanup), \$3.2/3.6-3.11 (semantic), \$5.1-5.5/5.8 (PDF-native), \$6.2 (IETM), \$7.1/7.4/7.5 (dedup, graph, cross-validation). o: \$4.6 dimension geom. Remaining o are the last vision pieces: \$4.5 callout-number OCR + \$4.8 symbol detection (next wave), and \$10.1 vision-language QA (GPU/host model). The self-standing repository is nearly feature-complete on the